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エジプト日本科学技術大学
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MSE 222

Fundamentals of Material Science (Lab)

Applications on bending, tension, and compression tests in daily life

Group 1 – Section 3

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During the lab we have encountered three different tests. These tests are bending test, tension test and compression test. We daily encounter each force of them. We will write some examples for these daily life applications.

1. **Bending test:**

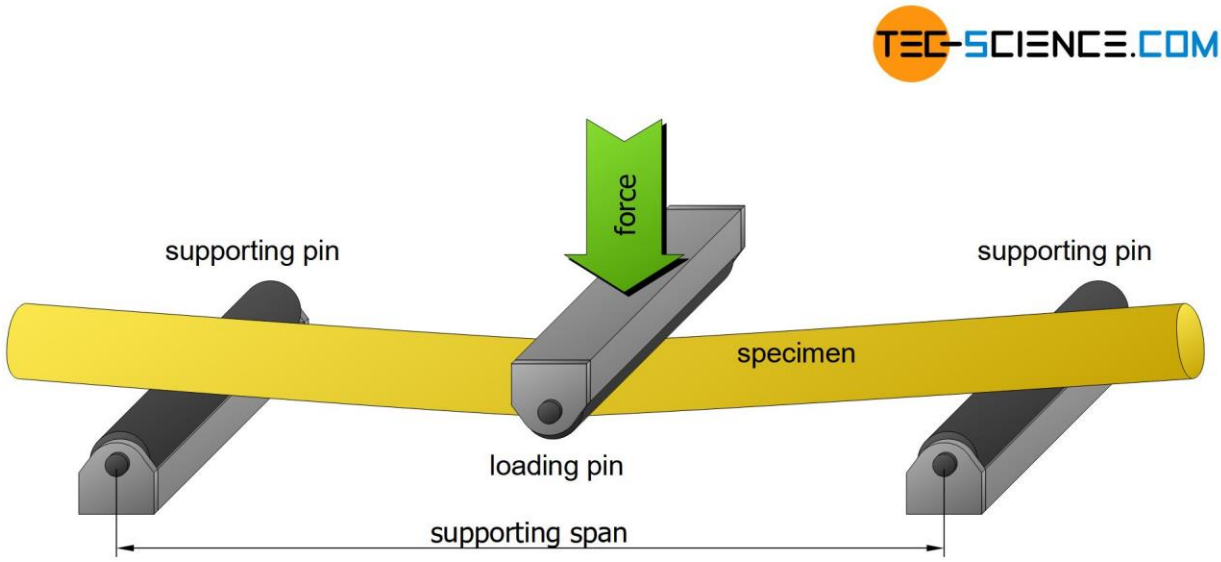


Figure 1 Bending test

We encounter the bending stress in many examples in our daily life. These examples are such as:

A) The tables in our homes experience a bending force.

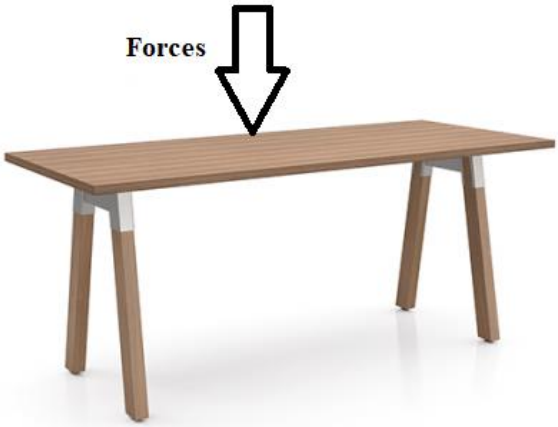


Figure 2 Bending forces on a table

B) The Bridge beams are experiencing a great bending moment de just to their self-weight. However, they are designed to not sag, and thus the bending moment is not visible.



Figure 3 Bending forces on a bridge beam

C) Bending of steel allows us to use it in construction.



Figure 4 Cut then bend steel bars

D) The cells of our homes experience a bending force from the loads on it.

E) The balconies of our rooms or our home experience bending forces.



Figure 5 Balcony experience bending forces

F) Wire balacing: the weight of the person is a bending force to the wire.

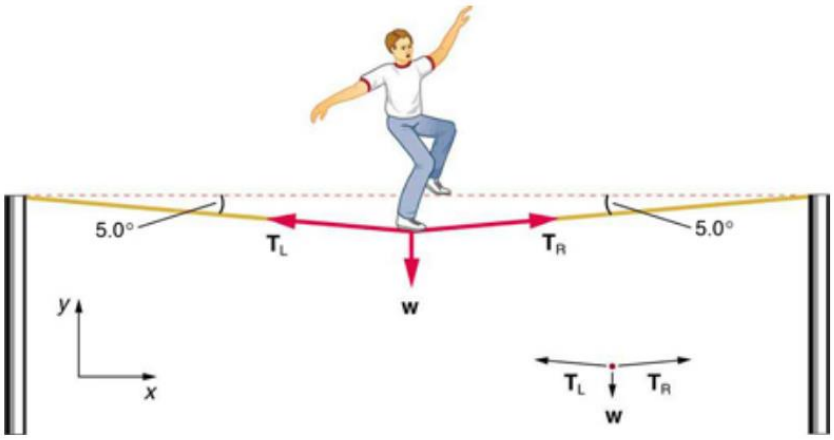


Figure 6 The wire encounter bending force

2. Tension test:

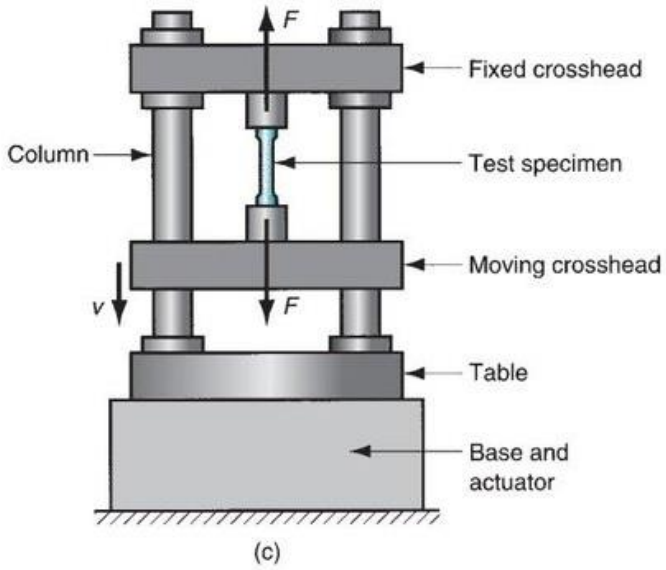


Figure 7 Tension Test

We encounter the Tension stresses in many examples in our daily life. These examples are such as:

A) Gym Equipment: wire experience tension force. This wire should have a tension test to choose the most suitable material



Figure 8 the wire experienced tension forces

B) Crane Machine:



Figure 9 Wires of crane machine experience tension forces

C) Car Towing:

The rod between the two cars experiences a tension force

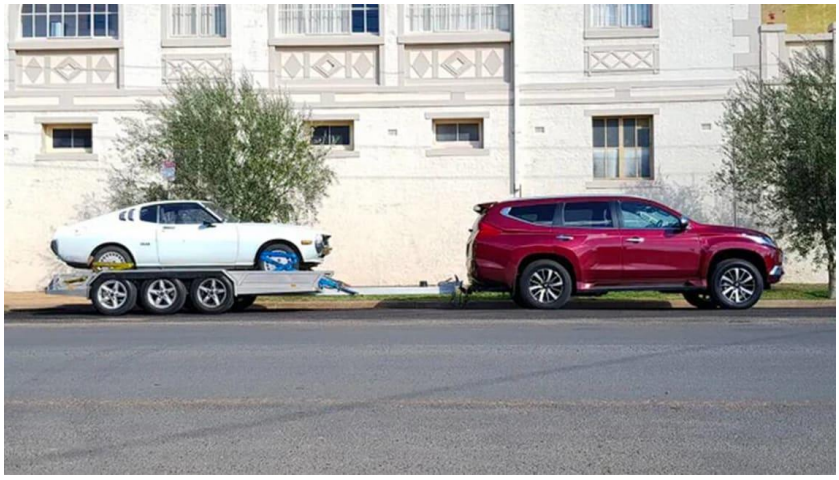


Figure 10 Tension force in car towing

D) Suspension Bridge:

In such a bridge, the number of cables kept hanging as they got pulled from both ends. The taunted cables produced the tension force which transmitted through the cables during the constant on boarding and offloading of vehicles on the bridge.



Figure 11 Suspension Bridge

E) Wire Balancing:

When a person starts walking on the wire, it stretches the wires downwards due to the weight. The stretched wires produced the tension force on wire that transmitted towards their both ends.

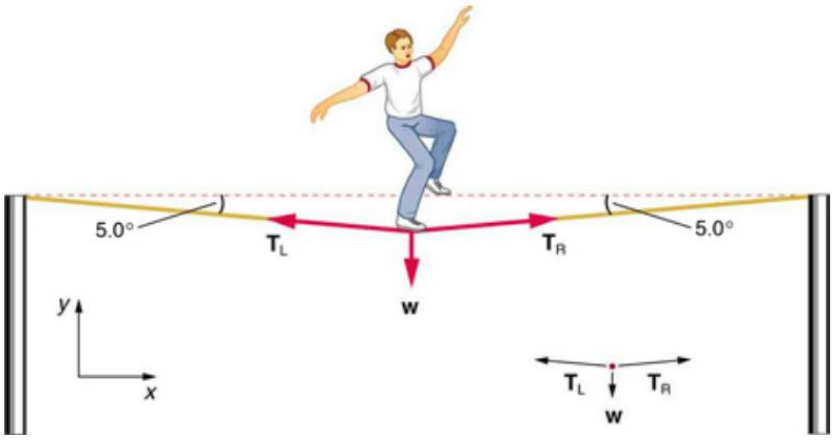


Figure 12 wire balancing

F) Elevators:

The wires in elevator experience tension forces.



Figure 13 Elevator wires

3. **compression test**

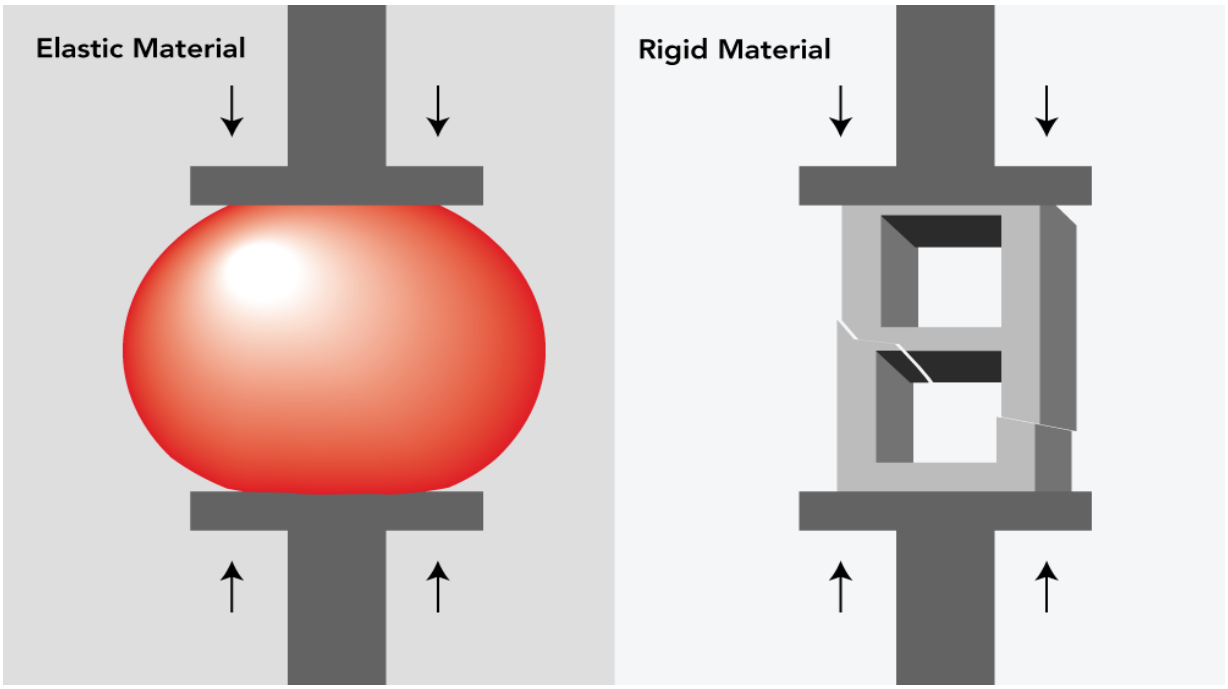


Figure 14: compression test

We encounter compression force in many examples in our daily life, theses examples are such as:

a) shoe sole:

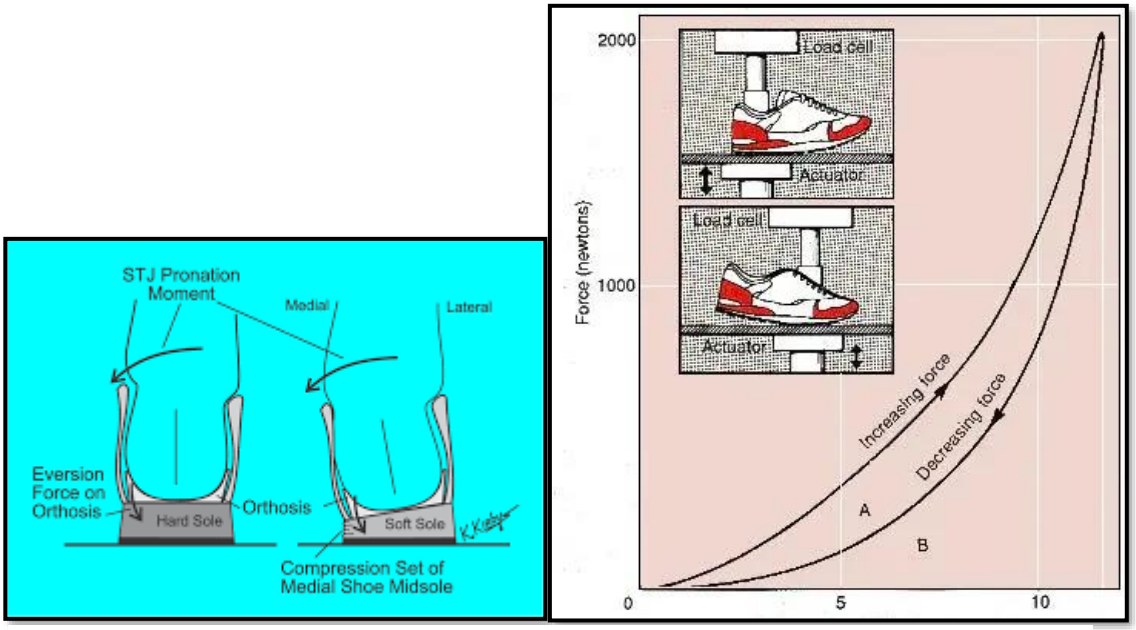


Figure 15: shoe sole compression forces

b) Bicycle Pump

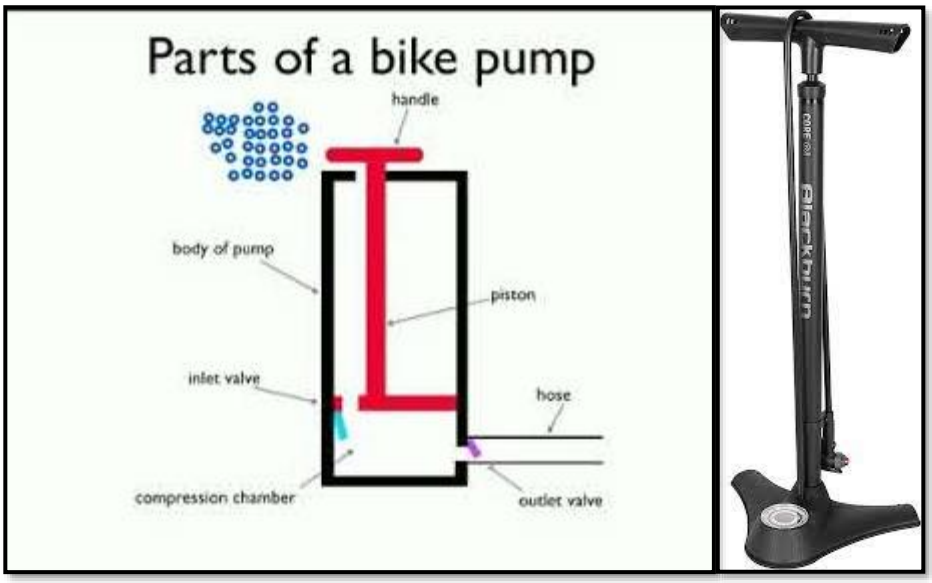


Figure 16 bicycle pump mechanism

c) springs

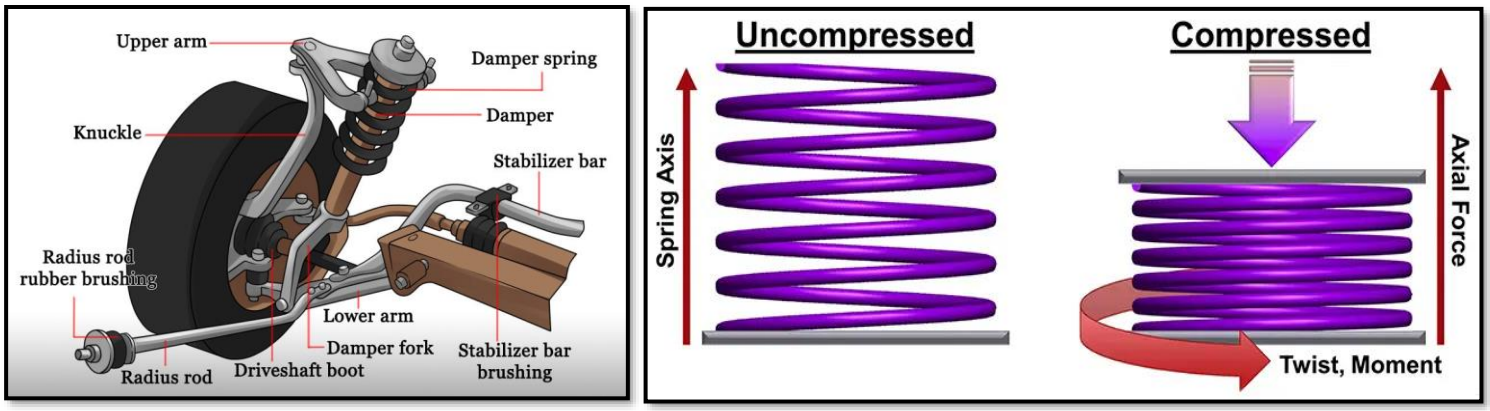


Figure 17 springs

d) bridges

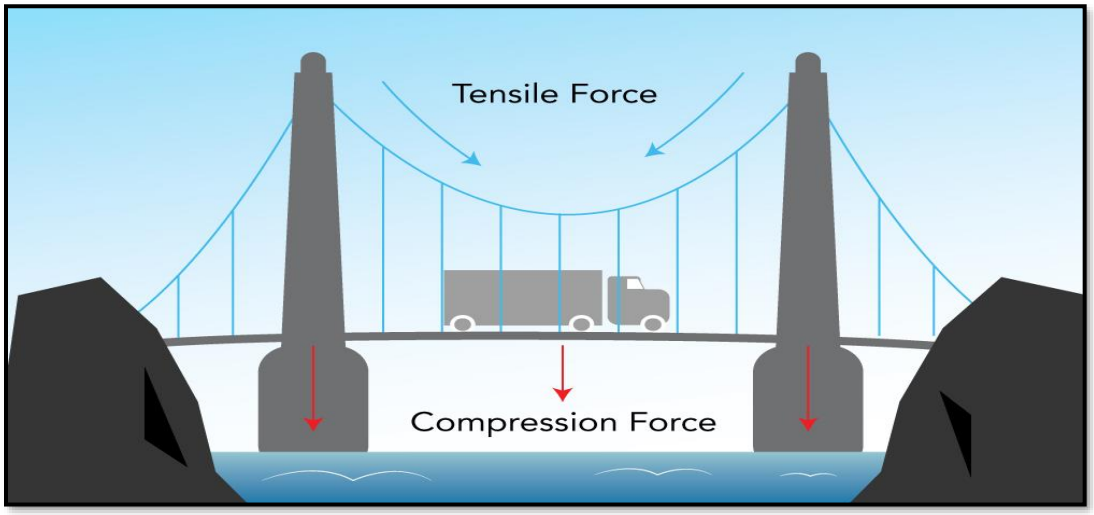


Figure 18 illustrate bridge forces

e) in many applications in the medical field as syringe

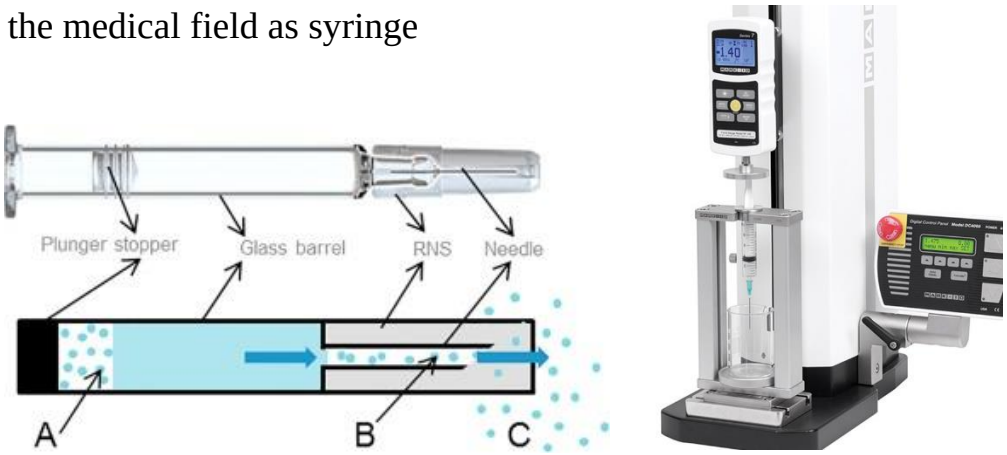


Figure 19 syringe compression mechanism